

To start, press CONTROL+PAGE DOWN (or press CONTROL+FN+DOWN ARROW if you're on a MacBook).

Take a tour

In just 10 steps, you'll be up and running with Excel, the world's most popular spreadsheet app.



Excel

top, press CONTROL+HOME. To start the tour, press CONTROL+PAGE DOWN. If you're on a MacBook, press CONTROL+FN+DOWN.

Let's go



A TIP FOR ZOOMING IN

Rest your cursor over the word "Take."
Then spread two fingers apart on your trackpad to zoom in. No trackpad? Hold down the Control key while you swipe forward on your mouse.

Ad

Here

1

2

3

4

5



Mc

In some
detail
text b

If the

Sum
follo

=ST

Here's

Sum t
follow

=ST

The fo

- A sin
cell re
- A ra
G48:C
- A cc

—

Mc

We al
sums

[NOTE](#)

=ST

NOTE
PivotT

—

[All ab](#)

Mc

[All ab](#)
[Excel](#)



Use numbers like a champ

Here are some ways to add up numbers in Excel:

- 1 Select the yellow cell under the amounts for fruit.
- 2 Type **=SUM(D4:D7)**, and then press Return. When you're done, the result is 170.
- 3 Now use a shortcut key. Select the yellow cell under the amounts for meat.
- 4 On the **Formulas** tab, click **AutoSum**. Then, press Return. The result is 140.
- 5 Now add only the numbers over 50. Select the last yellow cell. Type **=SUMIF(D11:D15,">50")** and then press Return. The result is 100.

Dive down for more detail

Next step

Fruit

Apples
Oranges
Bananas
Lemons

Item

Bread
Donuts
Cookies
Cakes
Pies

More about the SUM function

One of the above tips, we taught you how to use the SUM function. Here are more tips about it. Double-click a yellow cell on the right, and then read along with the video below.

If the SUM function could talk, it would say this:

Put up the wing:

...the values in cells D38, D39, D40, and D41.

=SUM(D38:D41)

Fruit

Apples
Oranges
Bananas
Lemons

is another way it can be used:

Item
Table

he
ing:

...the
value in
cell
D48...

...the values
in cells G48,
G49, G50,
and G51...

...and
100.

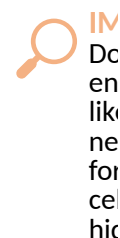
SUM (D48 , G48 : G51 , 100)

Formula above uses the following:

Single **cell reference**, which is the "address" or "name" of a cell. D48 is the single reference in the formula above.

Range of cells, which is a series of cells starting at one cell and ending at another. G48:G51 is the range of cells in the formula.

Constant. The constant in this formula is the number 100.



More about the SUMIF function

As we showed you the SUMIF function at the top of this sheet. The SUMIF function adds up totals based on a criterion. If the SUMIF function could talk, it would say this:

Item
Bread
Donuts
Cookies
Cakes
Pies

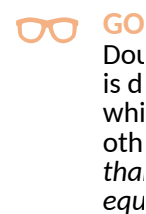
Sum up some
values based
on this
criterion:

....Look
through
these
cells...

...and if the
value is
greater than
50, sum it up.

SUMIF (D73 : D77 , ">50")

: If you find you are making a lot of SUMIF formulas, you might find that a PivotTable is a better solution. [See the PivotTable worksheet for more information.](#)



For more information on the web



[All about the SUM function](#)



[All about the SUMIF function](#)



[Excel training online](#)



Back to top

Next step



Amount
50
20
60
40

Meat	Amount
Beef	50
Chicken	30
Pork	10
Fish	50

Amount
50
100
40
50
20

Item	Amount
Bread	50
Donuts	100
Cookies	40
Cakes	50
Pies	20



EXTRA CREDIT

Try adding another SUMIF formula here, but add amounts that are *less than 100*. The result should be 160.

Amount
50
20
60
40
170



CHECK THIS OUT

Select these cells. Then in the lower-right corner of the Excel window, look for this:

Sum: 170

That's just another way to quickly find a total.

Amount	Item	Amount
20	Cars	20
	Trucks	10
	Bikes	10
	Skates	40

Total:
200

IMPORTANT DETAIL

Double-click this cell. You'll notice the 100 toward the d. Although it's possible to put numbers in a formula e this, we don't recommend it unless it's absolutely cessary. This is known as a **constant**, and it's easy to get that it's there. We recommend referring to another ll instead, like cell D16. That way it's easily seen and not lden inside a formula.

Amount	Item	Amount
50	Bread	50
100	Donuts	100
40	Cookies	40
50	Cakes	50
20	Pies	20
100		200

OD TO KNOW

Double-click this cell and you'll see that the formula is different. Specifically, the sum criteria is ">=50" which means greater than or equal to 50. There are other operators you can use like "<=50" which is *less than or equal to* 50. And there's "<>50" which is *not equals* 50.

Sav

Here's

1

2

3

4



Use

Some
copy v

1

2

Fill

Excel
in one

1

2

3

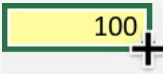
4

Mc



Save time by filling cells automatically

Here's how to use the fill handle in Excel:

- 1 Click the yellow cell with the number **100**.
- 2 Rest your cursor on the lower-right corner of the cell until it becomes a cross: 
- 3 Click the cross and drag down three cells. Excel will automatically fill the cells with the totals: **110**, **120**, and **130**. People call this "filling down."
- 4 Click the yellow cell with **200**, and fill again but this time drag the fill handle to the *right* to fill the cells. This is known as "filling right."

This:
50
50
50
50

This:
50
50
50
50
200

Dive down for more detail

Next step

Use the fill handle to copy cells

Sometimes you don't need the numbers to change as you fill. Instead, you just want to copy values to other adjacent cells. Here's how to do that:

- 1 Click the cell with the word **Produce**. Rest your cursor on the lower-right corner of the cell until it becomes a cross, then drag down three cells.
- 2 Now select the cell with the word **Fruit**. Rest your cursor on the lower-right corner again, and when you get the cross, double-click. That's another way to fill down in case you ever need to fill a long column.

Dept.
Produce

a series

can automatically fill some cells based on a series. For example, you can type Jan in the first cell, and then fill the other cells with Feb, Mar, etc.

Click the cell with the word **Jan**.

Rest your cursor on the lower-right corner of the cell until it becomes a cross, then drag right two cells. Excel detects a series, and fills in **Feb** and **Mar** for you.

Now select the cell with **Week 1**.

Rest your cursor on the lower-right corner again, and when you get the cross, *double-click it*.

Week 1



Intervals

15



EXP
Select
right
and
Janu
happ

More information on the web

[Copy a formula with the fill handle](#)

Back to top

Next step

Plus this:	Equals:	Plus this:	Equals:
50	100	75	175
60		75	
70		75	
80		75	



EXTRA CREDIT

Click and drag to select these four cells, and then press Control+D. That's the shortcut key for filling down. Can you guess what the shortcut for filling *right* is?

Plus this:	Equals:	Plus this:	Equals:
50	100	75	175
60	110	75	185
70	120	75	195
80	130	75	205

Category	Product	Count
Fruit	Apple	100
	Orange	200
	Banana	50
	Pears	100

Jan			
35	44	79	
74	64	56	
82	50	83	
90	22	89	

Qrtr 1

IMPORTANT DETAIL

Select this cell, and then drag the fill handle down 3 cells. After that, click blue button with the cross that appears. This is the **AutoFill Options** button, and it lets you change the fill immediately after. Choose another option like Copy cells or Fill Formatting Only. You never know when these might come in handy someday.

30

EXPERIMENT

Click these two cells, and then drag the fill handle to the right. Excel fills the series in increments of 15. Try changing 15 to 30 to other values, like 1 and 1.8. Or, Mon and Wed. Or, January and March. And then fill to the right again... see what happens!

Data

If you
delim

1

2

3

4

5



Spl

You m
updat
possib
infor
you're
diagra

1

2

3

4

5

—

Mc



—



[All ab](#)
[All ab](#)
[All ab](#)
[All ab](#)

Data stuffed into one column? Split it.

You want to split data, try **Text to Columns**. It lets you split a column based on a delimiter in the column. In this case, the commas are the delimiter.

- Click and drag to select the cells from **Nancy** all the way down to **Yvonne**.
- On the **Data** tab, click **Text to Columns**. Make sure that **Delimited** is selected, and then click **Next**.
- Under **Delimiters**, make sure that **Comma** is the only checkbox selected, and then click **Next**.
- Click the **General** option.
- Finally, click inside the **Destination** box and type **\$D\$7**. Then click **Finish** and **OK**.

Dive down for more detail

Next step

Split a column with formulas

You might want to write a formula to split data. This way, if the original data gets updated, then the split data will get updated as well. This is more advanced. But it is possible when using a handful of functions: LEFT, RIGHT, FIND, and LEN. For more information on each of these functions, see the links at the bottom of this sheet. But if you're curious, here's how we split cell C33. Make sure to follow along with the diagram on the right as you go through these steps:

- Double-click the yellow cell with **Yvonne**. We used the LEFT function to extract characters from the left side of cell C33. And to specify the number of characters to extract, we used the FIND function. Read the **How it works** diagram, and then press ESC when you're done.
- Then we created a **[Helper column]**. This was just to "help" extract the other text in the cell. It's meant to be temporary and could be hidden later.

HOW

Ex
tr
ac
t
ch
ar
ac
te
r

Double-click **Francis McKay** in the [Helper column]. You'll see that we used the RIGHT, LEN, and FIND functions to extract characters from the first space, up until the end of the cell.

Double-click **Francis**. Here we used almost the same formula as in step 1, but instead of extracting characters from C33, it extracts them from F33.

Double-click **McKay**. This is the same formula as in step 3, but it extracts characters from F33 instead of cell C33.

More information on the web

[All about the LEFT function](#)

[All about the RIGHT function](#)

[All about the FIND function](#)

[All about the LEN function](#)

Back to top

Next step

LEFT (C33, FIND (" ", C33) - 1) = RIGHT (C33, LEN (C33) - F1

...and find
the
character
position
number
of...

... the
first
space...

...in
this
cell.

...then
subtract 1
to exclude
the space
itself.

...and get the count
of characters
(character length)
of...

...t
his
cel
l...

...and
subtr
act
this:

Last name
McKay

s many characters. To specify the number of
he LEN function...

100 / " " 0000

ND (" " , C33))

Find the
character
position
number
of...

...the
first
space...

...in
this
cell.

Sw

Where

1

2

3

4

5



Tra

Some
form

1

2

3

4

5

Wh

An an
exam
functi

You a
Contr
Excel
the se
Excel

[All ab](#)
[Creat](#)

Mc



Switch data around by transposing it

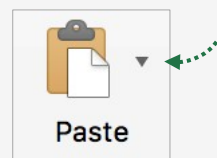
When you need to rotate columns and rows, you *transpose* them in Excel.

1 Click and drag to select the two rows of cells from **Item**, to **20**.

2 Now you'll copy the cells. Press Control C

3 Click the yellow cell.

4 On the **Home** tab, click the arrow next to the **Paste** button.

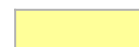


5 Click **Paste Special**, and then at the bottom, click the checkbox for **Transpose**. Click **OK**.

[Dive down for more detail](#)

[Next step](#)

Item
Amount



Transpose with a formula

Sometimes you don't want to copy and paste to transpose. In this case, you can use a formula to transpose rows and columns. Here's how to do that:

1 To transpose this data, you need to select some blank cells first. Since the data on the right has 6 columns and 2 rows, you need to select the opposite: 2 columns and 6 rows. Do this by selecting the yellow cells.

2 This is kind of tricky, so pay close attention. With those cells *still selected*, type the following: **=TRANSPOSE(C33:H34)**but don't press *Return*.

3 Press Control Shift Return

Item
Amount

So select

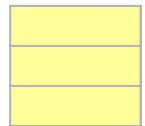


If you get #VALUE! as a result, try again starting at step 1.

Click any of the yellow cells to select just one. Look at the formula at the top of Excel. You'll see that the formula looks like this:

```
{=TRANSPOSE(C33:H34)}
```

Click another yellow cell. Look at the formula bar again. The formula is the same. Why? Because this is an **array formula**.



KEY
The
for

1) /
sel
mu

2) \
Col

3) c
arr
cell
arr
rec

What's an array formula?

An array formula can perform calculations on more than one cell in an array. In the example above, the array is the original data set in cells C33:H34. The TRANSPOSE function then switches the horizontal orientation of the cells to a vertical orientation.

Always finish an array formula with Control+Shift+Return, not just Return. Pressing Control+Shift+Return calculates the function against the array. When you're done, Excel puts special brackets { } around the formula. These brackets are a visual clue that the selected cell is part of an array formula. You can't type these brackets yourself. Excel puts them in when you press Control+Shift+Return.



EXC
Son
bec
Ent

More information on the web

[All about the TRANSPOSE function](#)

[Create an array formula](#)

 Back to top

Next step 

Bread	Donuts	Cookies	Cakes	Pies	
50	100	40	50	20	



EXPERT TIP
The shortcut key for
Paste Special is
Control+Option+V.

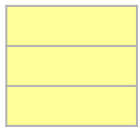
This data has 6

Bread	Donuts	Cookies	Cakes	Pies	
50	100	40	50	20	

...and 2 rows.

these 2 columns...

...and these 6 rows



before you type the formula.

EP IN MIND...

There are three things to keep in mind when using an array formula:

Always select multiple cells first, and then with those cells selected, start typing the array formula. That's the key: Select multiple cells first, then start typing.

When you're done typing an array formula, press **Control+Shift+Return**.

Once you enter an array formula, you cannot interrupt that new array. For example, you cannot type over or delete just one of the cells. You also cannot insert a new row or column within that array. If you need to do that, select all of the cells that have the array formula, press **Delete**, and then make your changes and reenter the formula.

LET SPEAK

Some people call array formulas "CSE formulas." That's because on a Windows computer, the **Return** key is called the **Enter** key.

Sort

1

2

3

4

5

Sort

There
time

1

2

Mc

Many
certai
availa

1

2

[More](#)
[More](#)

Mc



Sort and filter with ease

Let's say you want the departments in alphabetical order. Click in the **Department** column, and then click **Home > Sort & Filter > Sort A to Z**.

Sort December's amounts from largest to smallest. Click any cell in the **Dec** column, and then click **Home > Sort & Filter > Sort Largest to Smallest**.

Now you'll filter the data so that only the Bakery rows appear. Click **Home > Sort & Filter > Filter**. Filter buttons appear on the top row.

On the **Department** cell, click the filter button and then click to clear the **Select All** checkbox. Then, click to select **Bakery**.

Now clear the filter by clicking **Clear Filter** at the bottom of the Department filter pane, and then close the pane.

Dive down for more detail



Next step



Sort by date, or by color even

There are many ways to sort in Excel. Here are just two more ways to sort, but this you'll use the right-click menu:

You want the dates in order. So, press the Control key and then click a date and then click **Sort > Sort Oldest to Newest**. The rows get sorted in ascending date order by the Expense date.

Someone filled three cells with yellow. You can sort the rows by that color. Control-click a yellow cell, and then click **Sort > Put Selected Cell Color on Top**.

More ways to filter data

Some people type formulas to find amounts that are above average, or greater than a certain amount. But there's no need to type formulas when special filters are available.

1 On the **Hotel** cell, click the filter button and then click **Choose One > Above Average**. Excel calculates the average amount of the Hotel column, and then shows only rows with amounts greater than that average.

2 Now add a second filter. On the **Food** cell, click the filter button and then click **Choose One > Greater than...**, and then type **25**. Of the three rows that were filtered for above average, Excel shows two rows with Food amounts greater than 25. Click **Clear Filter** when you're done.

More information on the web

 [More information on sorting](#)

 [More information on filtering](#)


Back to top

Next step 

Department	Category	Oct	Nov	Dec
Meat	Beef	\$90,000	\$110,000	\$120,000
Bakery	Desserts	\$25,000	\$80,000	\$120,000
Produce	Fruit	\$10,000	\$30,000	\$40,000
Produce	Veggies	\$30,000	\$80,000	\$30,000
Deli	Sandwiches	\$90,000	\$35,000	\$25,000
Meat	Chicken	\$75,000	\$82,000	\$2,000,000
Bakery	Breads	\$30,000	\$15,000	\$20,000
Deli	Salads	\$80,000	\$40,000	\$20,000



EXTRA CREDIT

When you're done with step 5, try sorting alphabetically by two columns. Here's how: First sort **Department** alphabetically (that's step 1 on the left). Then click **Home > Sort & Filter > Custom Sort**. Add a second level for **Category**. After you click OK, **Department** will be sorted, and within each department, **Category** rows will be sorted in alphabetical order as well.

Expense date	Employee	Food	Hotel
4/4/2026	Jackie	\$21	\$3,820
4/3/2026	Mark	\$62	\$2,112
3/31/2026	Dave	\$25	\$1,611
4/6/2026	Tricia	\$30	\$3,085
4/2/2026	Jeff	\$69	\$528
4/1/2026	Laura	\$45	\$5,050



IMPORTANT DETAIL

You can't clear a sort order like you can a filter. So if you don't want your sort to stick, undo it by pressing Control+Z.

Expense date	Employee	Food	Hotel
4/4/2026	Jackie	\$21	\$3,820
4/3/2026	Mark	\$62	\$2,112
4/6/2026	Tricia	\$30	\$3,085
3/31/2026	Dave	\$25	\$1,611
4/1/2026	Laura	\$45	\$5,050
4/2/2026	Jeff	\$69	\$528

Task

A tab

1

2

3

4

5



Calc

One e
formu

1

2

3

4



Tot

Anotl
can n

1

2

3

4

5

6



[Creat](#)
[Using](#)
[Use c](#)

Mc



Tables make things a lot easier

Excel gives you special features and conveniences. Here's how to create one:

- 1 Click inside the data to the right, and then click **Insert > Table > OK**.
- 2 Now you have a table, which is a collection of cells that has special features. For starters: A table gives you banded rows for easier reading.
- 3 You can also create new rows easily. In the empty cell under **Meat**, type some text and then press Return. A new row for the table appears.
- 4 You can also create columns easily: In the lower-right corner of the table, click the small blue resize handle and drag it to the right 2 columns.
- 5 Notice how the two columns are created, formatted, and the text **Jan** and **Feb** are filled for you.

Dive down for more detail

Next step

Calculated columns in tables

One example of a convenience that tables give you: **calculated columns**. You type a formula once, and it gets automatically filled down for you. Here's how it works:

- 1 Select the cell under **Total**.
- 2 On the **Formulas** tab, click **AutoSum**. Then, press Return.
- 3 The first row of data gets summed up.
- 4 Very soon after, the same formula gets filled down for you so that you don't

have to do it yourself.

al rows in tables

her convenience in tables are **total rows**. Instead of typing a SUM formula, Excel make that total for you with a flip of a switch:

Select any cell within the table on the right.

At the top of the Excel window, the **Table** tab will appear.

On that tab, click **Total Row**.

The total of **\$24,000** is added to the bottom of the table.

But what if you wanted to know the average? Click the cell with **\$24,000**.

Click the down arrow and then click **Average**. The average amount of **\$3,000** appears.

ore information on the web

[Create a table](#)

[Using structured references in a table](#)

 Back to top

Next step 

Department	Category	Oct	Nov	Dec
Produce	Veggies	30000	80000	30000
Produce	Fruit	10000	30000	40000
Bakery	Breads	30000	15000	20000
Bakery	Desserts	25000	80000	120000
Deli	Sandwich	80000	40000	20000
Deli	Salads	90000	35000	25000
Meat	Beef	90000	110000	200000
Meat	Chicken	75000	82000	150000



EXPERT TIP

The shortcut key for inserting a table is Control+T.

Department	Category	Oct	Nov	Dec	Total
Produce	Veggies	\$30,000	\$80,000	\$30,000	
Produce	Fruit	\$10,000	\$30,000	\$40,000	
Bakery	Breads	\$30,000	\$15,000	\$20,000	
Bakery	Desserts	\$25,000	\$80,000	\$120,000	
Deli	Sandwiches	\$80,000	\$40,000	\$20,000	
Deli	Salads	\$90,000	\$35,000	\$25,000	
Meat	Beef	\$90,000	\$110,000	\$200,000	
Meat	Chicken	\$75,000	\$82,000	\$150,000	



EXPERIMENT

After putting in the calculated column, try typing over one of the cells in the column. What happens? If you see a green triangle, click it and then click the exclamation mark. You'll see that Excel's watchin' out for ya...

Department	Category	Sales
Produce	Veggies	\$1,000
Produce	Fruit	\$2,000
Bakery	Breads	\$3,000
Bakery	Desserts	\$1,000
Deli	Sandwiches	\$2,000
Deli	Salads	\$3,000
Meat	Beef	\$4,000
Meat	Chicken	\$8,000



GOOD TO KNOW

There's a shortcut for showing and hiding the total row. Click inside the table, and then press Control+Shift+T.

Slic

The p

1

2

3

4

5



Ins

Slicer:
the ri

1

2

3

4



[More](#)

Mc



Slicers are another way to filter

Panel of blue buttons to the right of the table is a **slicer**.

- Notice that the data on the right contains different sales amounts for different departments and categories.
 - In the panel of buttons to the right, click the **Bakery** button. Excel filters out all rows except for the Bakery rows.
 - Click another department button, and it filters again.
 - Now hold down the Command key, and click just 2 of the buttons to see those two departments.
 - Click the Clear filter button at the top-right corner of the buttons. It's the one with the "x".
-

Dive down for more detail



Next step



Create a slicer

Slicers can be used when you first create a **table**. We did that already on the data to the right.

- Click anywhere in the table on the right.
- On the **Insert** tab, click **Slicer**.
- Select the check box for **Department**, and then click **OK**.
- Move the slicer buttons closer to your data, if necessary. And start slicing!

More information on the web

 [More on filtering and slicing](#)


Back to top

Next step



Department	Category	Sales
Produce	Veggies	\$30,000
Deli	Salads	\$90,000
Bakery	Breads	\$30,000
Meat	Chicken	\$75,000
Deli	Sandwich	\$80,000
Bakery	Cakes	\$25,000
Meat	Beef	\$90,000
Produce	Fruit	\$10,000
Produce	Nuts	\$22,000
Deli	Condiments	\$4,500
Bakery	Pies	\$30,000
Meat	Pork	\$75,000

This shape represents a table slicer. Table slicers are supported in Excel or later.

If the shape was modified in an earlier version of Excel, or if the workbook



EXPERT TIP

Slicers can also be used on PivotTables. You'll learn about PivotTables on Sheet 10.

Department	Category	Sales
Deli	Salads	\$90,000
Bakery	Breads	\$30,000
Deli	Sandwich	\$80,000
Bakery	Cakes	\$25,000
Deli	Condiments	\$4,500
Bakery	Pies	\$30,000
Produce	Veggies	\$30,000
Meat	Chicken	\$75,000
Meat	Beef	\$90,000
Produce	Fruit	\$10,000

Produce	Nuts	\$22,000
Meat	Pork	\$75,000



EXPERIMENT

After adding the slicer, make sure it is selected. Then at the top of Excel, click the **Slicer** tab. Play around with the different color styles. Or, resize the slicer panel to be wider, and set the columns to 2.

Ins

Drop-

1

2

3

4

5



Be

We ju
what
You'd
by cr

1

2

3

4

5

6

7

8

[Create](#)
[Add o](#)

Mc

Insert a drop-down list

Drop-down lists make data entry easier for people. Here's how to do one:

- 1 We want only three department names to be valid entries for each of the foods on the right. Those departments are Produce, Meat and Bakery.
- 2 Click and drag to select the yellow cells under **Department**.
- 3 On the **Data** tab, click **Data Validation**. Under **Allow**, click **List**.
- 4 In the **Source** box, type **Produce, Meat, Bakery**. Make sure to put commas in between them. Click **OK** when you're done.
- 5 Now click the yellow cell next to **Apples**, and you'll see a drop-down menu.

Dive down for more detail

Next step

Best practice for drop-downs: Use a table.

I just taught you how to insert a drop-down menu for the list of departments. But what if that list changes? For example, what if there is a new department called Dairy? I have to update the data validation dialog box. But there's a more efficient way of creating a table first:

- 1 In column F, click a cell with a department. For example, click **Meat**.
- 2 Create a table by pressing Control T and then **OK**.
- 3 Now you'll set up the data validation again. In column D, select all of the blank cells under **Department**.

On the **Data** tab, click **Data Validation**. Under **Allow**, click **List**.

Click inside the **Source** box, then click the arrow button to the right.

Click and drag to select just the **Produce**, **Meat** and **Bakery** cells in column F. Then click the down arrow button or press RETURN.

You should see this in the **Source** box: **=F\$32:F\$34**. (If you don't see that you can type it in.) Click **OK**.

Now click the drop-down arrow. There are only three departments: Produce, Meat and Bakery. But if you add a new department in column F under Bakery, it will get updated with the new department.

More information on the web

[Create a drop-down list](#)

[Add or remove items from a drop-down list](#)

 Back to top

Next step 

Food	Department
Apples	
Beef	
Bananas	
Lemons	
Broccoli	
Kale	
Ham	
Bread	
Chicken	
Cookies	
Cakes	
Pies	



GOOD TO KNOW

Drop-down lists help ensure people enter valid data. So it makes sense that drop-downs are a part of a larger group of features known as **data validation**.

There are other data validation methods. For example, you can restrict entry to whole numbers, dates, or even minimum and maximum amounts. There are many options available, and you can read more about them by clicking the link at the bottom of this sheet.

Food	Department
Apples	
Beef	
Bananas	
Lemons	
Broccoli	
Kale	
Ham	
Bread	
Chicken	
Cookies	
Cakes	

Department
Produce
Meat
Bakery



EXPERT TIP

Often people put their validation lists like this out of the way on another sheet. That way others won't be tempted to change the list.

Gre

1

2

3

4

5



Ho

In sch
two a

In Exc

- The
- The

Each

- A va
dollar
value
- A ca
horiz



Sec

You c
axis t

A pop
addit
Some
That'
callec
the b



[Creat](#)
[Creat](#)

Mc



Best charts recommended for you

Click anywhere in the data to the right, and then click **Insert > Recommended Charts**.

You'll see several recommendations. Click the **Clustered Column** option.

A column chart appears showing total number of conference attendees per year. Feel free to move it anywhere you'd like.

Now you'll add a trendline. Select the chart, and the **Chart Design** tab will appear at the top of the Excel window.

On the **Chart Design** tab, click **Add chart element > Trendline > Linear**. Now you have a trendline that shows the general direction of the attendance over time.

Dive down for more detail

Next step

Horizontal and vertical axes

Now you might have learned that there is an x-axis and a y-axis. Excel has these axes as well, but it calls them something different.

Excel this is what they are called:

The x-axis along the bottom is called the **horizontal axis**.

The y-axis that runs up and down is called the **vertical axis**.

An axis can either be a value axis or a category axis.

A **value axis** represents numerical values. For example, a value axis can represent years, hours, duration, temperature, and so on. The vertical axis on the right is a value axis.

A **category axis** represents things like dates, people names, product names. The horizontal axis on the left has years so this is a category axis.

Vertical
axis

Secondary axis

You can also use a **secondary axis** in a chart. A secondary axis is an additional value axis that can show different values than the other value axis.

A popular example is on the right. It's the same as the chart above, but it has an additional secondary vertical axis that represents the sales amounts for each month. We would say that by having a secondary axis, you almost have "two charts in one." This is true. This chart is both a column chart and a line chart. These kind of charts are called **Combo charts** in Excel. If you're interested in this kind of chart, click the link at the bottom of this sheet.

More information on the web

 [Create a chart](#)

 [Create a combo chart](#)


[Back to top](#)

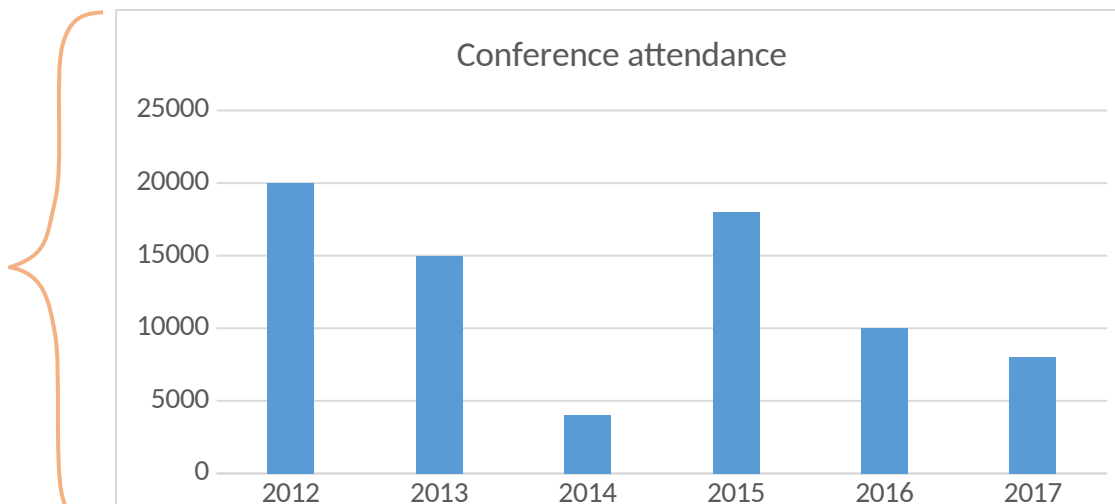
[Next step](#) 

Date	Conference attendance
2012	20000
2013	15000
2014	4000
2015	18000
2016	10000
2017	8000

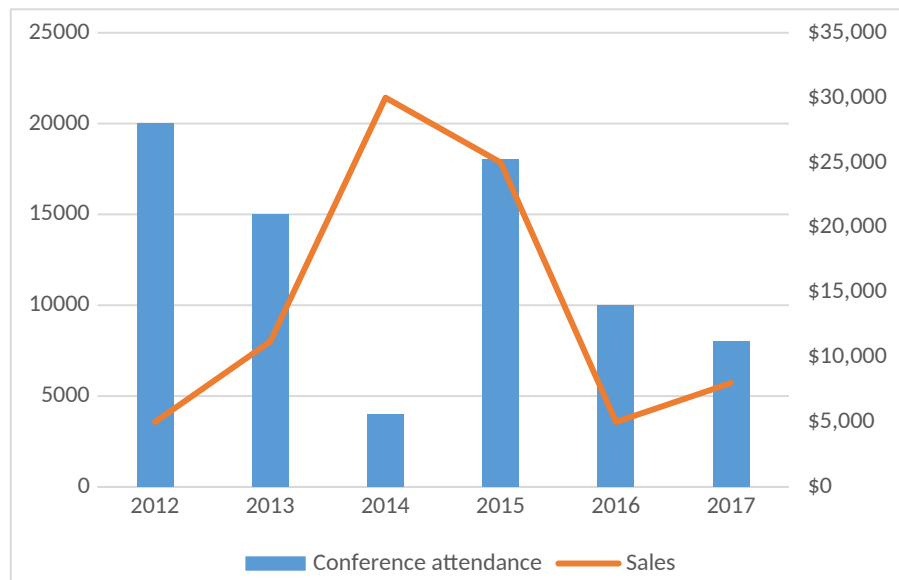


EXTRA CREDIT

Want a data table directly under the chart? Click the chart. On the **Chart Design** tab, click **Add Chart Element > Data Table > With Legend Keys**.



Horizontal axis
(Category axis)



Secondary axis

Date	Conference attendance	Sales
2012	20000	\$5,000
2013	15000	\$11,200
2014	4000	\$30,000
2015	18000	\$25,000
2016	10000	\$5,000
2017	8000	\$8,000

Data table



EXTRA CREDIT

Try making a combo chart. Click inside the data above, and then click **Insert > Recommended Charts**. Choose the **Clustered column** option.

Then, click one of the orange columns. At the top, click the **Chart Design** tab. On that tab, click **Change Chart Type** and choose the first **Line** option.

Finally, click the orange line, and then click the **Format** tab at the top. Select **Format Pane**. Then click **Secondary axis**.

ondary
axis

ia that supports
: secondary axis
above

Sur

1

2

3

4

5



Cre

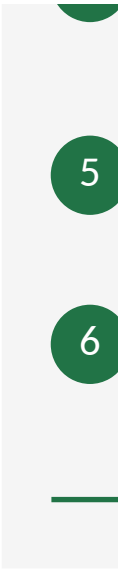
Now y
to sur

1

2

3

4



[Create](#)
[Sort a](#)



Summarize data with PivotTables

Look at the Date, Salesperson, Product and Amount columns. Can you quickly identify which product is the most profitable? Or which salesperson is the leading seller? That's where the PivotTable below can help.

When we created the PivotTable, we clicked a few buttons so that the data could be summarized. Now we know which product is the most profitable.

Now you'll pivot the data so that you can find out which salesperson is the leading seller. Click any cell inside the PivotTable, and then on the **PivotTable Analyze** tab, click **Field List**.

A pane appears. At the top of the pane, uncheck **Product**.

At the top of the pane, click the checkbox for **Salesperson**. Now you can see who's the leading salesperson.

Dive down for more detail



Next step



Create a PivotTable

You'll create the PivotTable yourself so that you know how to make one when you need to summarize data.

Click a cell inside the data on the right, and then on the **Insert** tab, click **PivotTable**.

In the dialog that appears, click **Existing Worksheet**, and then type C42 in the **Table/Range** box. Click **OK**.

A pane appears on the right.

At the top of the pane, click the checkbox for **Amount**. When you do that, the Amount field will get added to the Values area at the bottom of the

the Amount field will get added to the values area at the bottom of the pane. And, at the same time the amounts are totaled for each product in the PivotTable.

At the top of the pane, click the checkbox for **Salesperson**. When you do that, the Product field gets added to the Rows area at the bottom of the pane. And, the product data appears as **Row labels** in the new PivotTable.

Congratulations, you made a PivotTable. But there is a lot more you can do. So click the link at the bottom of this sheet if you want to learn more.

More information on the web

 [Create a PivotTable to analyze worksheet data](#)

 [Sort a PivotTable](#)

 Back to top

Next step 

Date	Salesperson	Product	Amount
2/8/2026	Anne	Beer	\$ 1,400
2/13/2026	Mark	Wine	\$ 1,010
3/2/2026	Anne	Beer	\$ 750
3/6/2026	Mark	Soda	\$ 510
3/26/2026	Mariya	Soda	\$ 1,600
4/6/2026	Laura	Wine	\$ 680

PivotTable



Product	Sum of Amount
Beer	\$2,150
Soda	\$2,110
Wine	\$1,690
Total Result	\$5,950

Date	Salesperson	Product	Amount
2/8/2026	Anne	Beer	\$ 1,400
2/13/2026	Mark	Wine	\$ 1,010
3/2/2026	Anne	Beer	\$ 750
3/6/2026	Mark	Soda	\$ 510
3/26/2026	Mariya	Soda	\$ 1,600
4/6/2026	Laura	Wine	\$ 680

More questions about Excel?

Keep going! There is more to learn!

[LinkedIn Learning: Video courses for all levels—from beginner to advanced. Take at your own pace.](#)



LinkedIn Learning

Video courses for all levels—from beginner to advanced. Take at your own pace.

[Learn more](#)



Community

Ask questions and connect with other Excel fans at the Excel Community.

[Learn more](#)



What else is new?

Office 365 subscribers get continual updates and new features. Check them out here.

[Learn more](#)

[Community: Ask questions and connect with other Excel fans.](#)

[What else is new? Office 365 subscribers get continual updates and new features.](#)